Town Hall: A 100 MW AI Data Center

Day 4: Infrastructure, Cost Allocation, and Local Consent

Scenario. The fictional city of Riverton must decide whether to approve a 100 MW AI data center. The project is economically attractive but would arrive faster than planned generation and grid upgrades. All figures below are stylized but plausible and are designed for analysis rather than prediction.

Decision

The City Council must choose: **approve**, **approve with conditions**, **delay**, or **reject**. Conditional approval must specify who pays, what performance is required, and how compliance will be monitored.

Common Factsheet

Project	100 MW initial load, with a contractual option to expand to 180 MW after four years. At full initial use, annual electricity consumption could approach 876 GWh.
Grid	Regional reserve margin is 11%. A new substation and transmission reinforcement are estimated at $75–140 million. Standard interconnection is projected to take four to six years. A faster non-firm connection is possible if the center accepts curtailment during tight conditions.
Power proposal	The developer proposes grid power plus a renewable PPA, batteries, and temporary onsite gas generation. The PPA matches annual consumption but not necessarily hourly use.
Water and heat	Estimated water use is 250,000–450,000 gallons per day, depending on cooling design and weather. Summer restrictions already occur in two to four weeks of a typical year.
Economy	1,100 construction jobs for two years; 45–65 permanent jobs; projected local tax revenue of $5–8 million annually. The developer requests property-tax relief worth $60 million over ten years.
Community	The site is 350 yards from the nearest neighborhood. Concerns include noise, diesel-generator testing, construction traffic, water use, and electricity rates.
Cost allocation	The utility proposes a special tariff but has not resolved how much of the grid upgrade would be recovered from the developer versus other customers.
Potential flexibility	The center could install batteries, accept non-firm service, participate in demand response, and disclose hourly load forecasts. These commitments are negotiable.

Roles

- AI data-center developer
- Electric utility and regional grid operator
- City Council and mayor's office
- Residential ratepayers and nearby residents
- Environmental and water coalition
- Labor and regional economic-development coalition
- Major local institutions: hospital, university, and small-business association

Preparation (10 minutes)

Each group must identify its preferred decision, three arguments, two questions for other actors, and at least two concessions it could accept.

Town Hall (25 minutes)

1. One-minute opening statement from each group.
2. Ten minutes of questioning and public negotiation.
3. Five minutes of private coalition building and revised offers.
4. Council decision and four-minute explanation.

Conditions the Council May Impose

- developer-funded grid upgrades or a minimum cost share;
- non-firm connection, curtailment, or demand-response obligations;
- hourly clean-power matching rather than annual accounting;
- water limits, reclaimed-water use, and drought restrictions;
- noise, generator-testing, and air-quality limits;
- community benefits, workforce commitments, and tax terms;
- public disclosure of load, water use, emissions, and outages;
- independent monitoring and penalties for noncompliance.

Debrief

Who receives the benefits? Who bears the infrastructure and reliability risks? Which conditions convert the project from a private investment into an acceptable public bargain?

Source note: the exercise reflects the policy issues emphasized in IEA, *Key Questions on Energy and AI* (2026), including grid bottlenecks, cost allocation, flexible connections, disclosure, and local social acceptance.